→ Tabela verdade

E_3	E_2	E_1	E_0	S_2	S_1	S_0
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	1	0	0
0	1	0	1	0	0	0
0	1	1	0	0	0	1
0	1	1	1	0	1	0
1	0	0	0	0	1	1
1	0	0	1	1	0	0
1	0	1	0	0	0	0
1	0	1	1	0	0	1
1	1	0	0	0	1	0
1	1	0	1	0	1	1
1	1	1	0	1	0	0
1	1	1	1	0	0	0

– Forma canônica usando $\sum_m$

$$S_2 = \sum m(4,9,14) = \bar{E}_3 E_2 \bar{E}_1 \bar{E}_0 + E_3 \bar{E}_2 \bar{E}_1 E_0 + E_3 E_2 E_1 \bar{E}_0$$

$$S_1 = \sum m(2,3,7,8,12,13) = \bar{E}_3 \bar{E}_2 E_1 \bar{E}_0 + \bar{E}_3 \bar{E}_2 E_1 E_0 + \bar{E}_3 E_2 E_1 \bar{E}_0 + E_3 \bar{E}_2 \bar{E}_1 \bar{E}_0 + E_3 E_2 \bar{E}_1 \bar{E}_0 + E_3 \bar{E}_2 \bar{E}_1 E_0$$

$$S_0 = \sum m(1,3,6,8,11,13) = \bar{E}_3 \bar{E}_2 \bar{E}_1 E_0 + \bar{E}_3 \bar{E}_2 E_1 E_0 + \bar{E}_3 E_2 E_1 \bar{E}_0 + E_3 \bar{E}_2 \bar{E}_1 \bar{E}_0 + E_3 \bar{E}_2 E_1 E_0 + E_3 E_2 \bar{E}_1 E_0$$

– Mapas de Karnaugh + forma simplificada POS

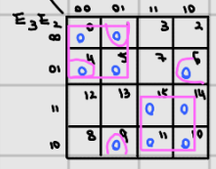
↳ S_2:



$$\bar{S}_2 = \bar{E}_3 \bar{E}_2 + \bar{E}_3 E_1 + \bar{E}_2 E_1 + E_2 E_0 + E_3 \bar{E}_1 \bar{E}_0$$

$$S_2 = (E_3 + E_2)(E_3 + \bar{E}_1)(E_2 + \bar{E}_1)(\bar{E}_2 + \bar{E}_0)(\bar{E}_3 + E_1 + E_0)$$

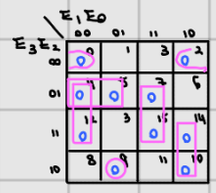
↳ S_1:



$\bar{S}_1 = \bar{E}_3\bar{E}_1 + E_3E_1 + \bar{E}_2\bar{E}_1E_0 + \bar{E}_3E_2\bar{E}_0$

$S_1 = (E_3 + E_1)(\bar{E}_3 + \bar{E}_1)(E_2 + E_1 + \bar{E}_0)(E_3 + \bar{E}_2 + E_0)$

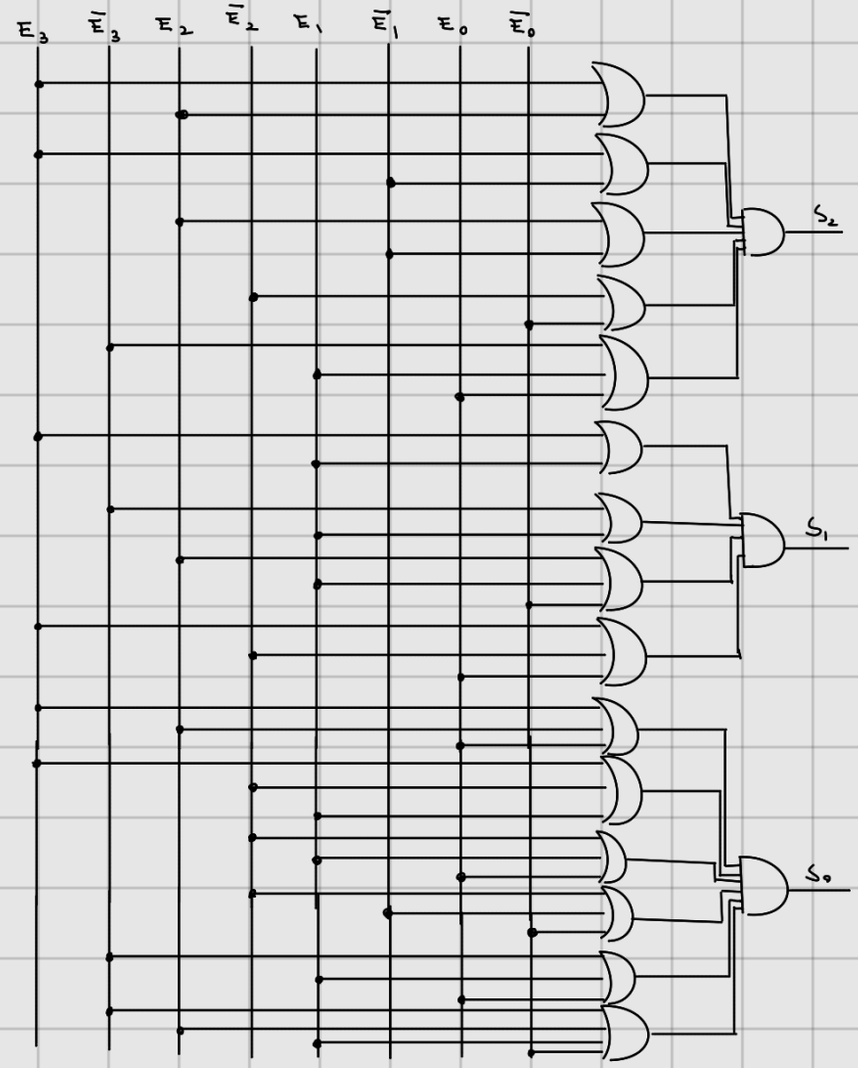
↳ S_0:



$\bar{S}_0 = \bar{E}_3\bar{E}_2\bar{E}_0 + \bar{E}_3E_2\bar{E}_1 + E_2\bar{E}_1\bar{E}_0 + E_2E_1E_0 + E_3E_1\bar{E}_0 + E_3\bar{E}_2\bar{E}_1E_0$

$S_0 = (E_3 + E_2 + E_0)(E_3 + \bar{E}_2 + E_1)(\bar{E}_2 + E_1 + E_0)(\bar{E}_2 + \bar{E}_1 + \bar{E}_0)(\bar{E}_3 + E_1 + E_0)(\bar{E}_3 + E_2 + E_1 + \bar{E}_0)$

- Diagrama do circuito



- Decisões de implementação do circuito: por ser um circuito relativamente simples, foram usados apenas componentes fundamentais da lógica combinatorial (portas lógicas OR, AND e inversores NOT).

- Visualização das formas de onda do caso de teste:

